Sliding Dual-Window-Inspired Reconstruction Network for Hyperspectral Anomaly Detection

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***Abstract*— Hyperspectral anomaly detection (HAD) aims to identify anomalous objects that deviate from surrounding backgrounds in an unlabeled hyperspectral image (HSI). Most available neural networks that make use of the reconstruction error to perform HAD tend to fit both backgrounds and anomalies, resulting in small reconstruction errors for both and not being effective in separating targets from background. To address this issue, we develop Dual-window-inspired reconstruction Network (DirectNet), a new background reconstruction network for HAD that seamlessly integrates a sliding dual-window model into a blind-block architecture. Concretely, DirectNet establishes an inner window within the network's receptive field by erasing the center block information so that the content of the inner window remains invisible during the reconstruction of the central pixel. In addition, the depth of our reconstruction network is adaptive to the size of the input image patch, ensuring that the network's receptive field aligns with the dimensions of the input patch. The receptive field outside the inner window is considered an outer window. This weakens the impact of anomalies on the reconstruction process, causing the reconstructed pixels to converge toward the background distribution in the outer window region. Consequently, the reconstructed HSI can be regarded as a pure background HSI, leading to further amplification of reconstruction errors for anomalous targets. This enhancement improves the discriminatory ability of DirectNet. Specifically, DirectNet solely utilizes the outer window information to predict/reconstruct the central pixel. As a result, when reconstructing pixels inside anomalous targets of different sizes, the targets primarily fall within the inner window. Comprehensive experiments (conducted on four datasets) demonstrate that DirectNet achieves competitive performance compared to other state-of-the-art detectors.**

***Index Terms*— Blind-spot network, convolutional neural networks (CNNs), deep learning (DL), hyperspectral images (HSIs), image reconstruction, self-supervised learning.**

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I. Introduction

HYPERSPECTRAL images (HSIs) can discriminate the fine spectral features of different ground materials [1], [2], and this property enables these images to be applied for detecting and locating objects of interest [3], [4]. Among many different HSI interpretation tasks, hyperspectral anomaly detection (HAD) is extensively utilized in both military and civilian applications by virtue of its ability to identify anomalous targets with no prior information [5], [6]. The anomalies are commonly defined as objects that occupy an extremely small area and deviate from the overall pattern of their surrounding background as well [7], [8]. The absence of prior knowledge about targets and background in the HAD task poses a critical challenge to separate anomalies from background and numerous detectors model such background through diverse approaches, where those samples not conforming to the background pattern are defined as anomalies. The mainstream existing methods are classified into three categories: statistical modeling-based, representation learning-based, and deep learning (DL)-based approaches.

A. Statistical Modeling-Based Methods

As the baseline of the statistical-based methods, the Reed–Xiaoli (RX) detector assumes that the background of an HSI obeys a multivariate Gaussian distribution and typically estimates the background using the global information of the image [9]. Those pixels with a large Mahalanobis distance from the estimated background have more potential to be considered as anomalies in a way that can be viewed as a global RX (GRX) detector. In order to alleviate the anomalous contamination during the process of background modeling, the local RX (LRX) [10] uses a sliding dual-window strategy. Here, the outer window pixels are analyzed to estimate the background, while the inner window avoids the leakage of target information. The weighted RX [11] method boosts the detection performance of the model by endowing the potential background with a high weight. To enhance the separability of anomalies from backgrounds, Kwon and Nasrabadi [12] adopted a kernel trick into the RX detector, and Tao et al. [13] introduced the fractional Fourier transform [14] into the HAD task. Subsequently, Li et al. [15] proposed the sliding dual-window-based 2S-GLRT method to locate multipixel anomalous targets of the HSI. Vincent et al. [16] modified the RX scheme by adopting the replacement model as a more realistic anomaly detection framework. Yang et al. [17] combined the detection results of different features on a random RX detector with the aid of ensemble learning to leverage the multiple characteristics in HSI.

B. Representation Learning-Based Methods

Since then, representation learning (encompassing sparse representation (SR) [18], [19], [20], [21], collaborative representation (CR) [22], and low-rank representation (LRR) [23], [24], [25]) has attracted extensive attention [26], [27], [28]. Among them, SR-based studies emphasize that normal/background samples can be represented by only a few atoms in an overcomplete dictionary, while anomalous ones cannot [29]. CR-based detectors [30], in contrast, focus on the collaborative relationships among all dictionary atoms and thus analyze whether each sample can be linearly represented by its surrounding neighbors. Both of them take reconstruction residuals to indicate the degree of anomaly of the testing pixels. To acquire more accurate detection results, many variants of CRD have been presented [22], [31]. Examining the global structural information of HSIs from the perspective of LRR, an HSI can be decomposed into background (with low-rank properties) and anomalies (with sparse properties) [32], [33]. Zhang et al. [34] employed the Mahalanobis distance between the pixels to be measured and the low-rank matrix to detect anomalies in the HSI, while Li et al. [35] introduced the Manhattan distance to recognize anomalies and represented the sparse matrix by utilizing a mixture of Gaussian noise models. In addition, LRR-based methods have emerged with diverse background dictionary constructions for better representation of HSIs, such as the works by Xu et al. [36] and Fu et al. [37], who obtained the background dictionary by leveraging K-means clustering, while Qu et al. [38] performed mean-shift clustering in the abundance matrix and separated anomalous pixels from the constructed dictionary. Given the spatial correlation of neighboring pixels, the total variation technique (which allows dictionary atoms to represent the background more efficiently) was introduced into LRR-based approaches [39]. Besides, HSI can be directly treated as a third-order tensor, and taking into account the use of both spatial and spectral information of HSI, He et al. [40] recently proposed a novel tensor low-rank approximation method for performing HAD.

C. DL-Based Methods

In recent years, DL has emerged as a successful approach in HSI processing, including the HAD task, by virtue of its powerful capabilities for mining deep characteristics [41], [42], [43]. The prevailing DL-based HAD methods can be categorized as similarity measurement models and reconstruction models.

Similarity measurement models commonly incorporate a convolutional neural network (CNN) [44] as the backbone, and a vast number of training pixel pairs are selected in advance from the labeled HSIs. The paired pixels from the same category are labeled as 0, while the pixels paired from different categories are labeled as 1. This value means the dissimilarity between the two. This property empowers the backbone network to measure the similarity of the input paired pixels, which is then transferred into the HAD task for identifying those anomalous pixels that are significantly different from their surroundings. Li et al. [45] considered the difference between the pixel pairs as the input, causing the constructed single-stream CNN to directly output the difference scores between the tested pixel and outer window pixels. Unlike this, Rao et al. [46] devised a Siamese network with feature extraction and similarity measurement abilities and took the original pixel pairs as the input to explore the similarity between each pixel and the surrounding ground objects at the latent layer. However, the detection efficiency of this family of methods is reliant on the quantity of paired data and the quality of labels, and transferability cannot be fundamentally guaranteed.

Conversely, reconstruction model-based methods have demonstrated their effectiveness and adaptability in the unsupervised HAD task, due to the fact that they directly apply unlabeled HSIs to be detected as training data and mine the intrinsic abstract features in unsupervised learning fashion. This category of methods primarily applies autoencoders [AEs or generative adversarial networks (GANs)] to reconstruct the raw data and employs the reconstruction error to obtain the final anomaly detection result. Since background pixels occupy the overwhelming majority of the HSI, the model could fully learn the background characteristics and emerge as a well-behaved background reconstructor. The proportion of anomalies is quite low, rendering the model to generate large reconstruction errors for them. Bati et al. [47] introduced AEs to the HAD task with a first attempt to reconstruct all samples in the detected image. To exploit the spatial features of HSIs in a reasonable way, Xiang et al. [48] introduced a guided model in the hidden layer of the AE to strengthen the network's power to represent the background, while Fan et al. [49] imposed a graph regularization constraint on the hidden features of the AE to enhance the robustness of the model in reconstructing background samples. Wang et al. [50] implemented an automatic end-to-end HAD, with the aid of a fully convolutional AE. Furthermore, unlike the traditional AE, the memory AE (with an external memory module recording latent representations of background samples) was introduced into HAD in [51], [52]. In order to inspire AEs to represent the background more prominently, several efforts have been dedicated to embedding discriminators into AEs (forming GANs) by also imposing Gaussian constraints on them [53] or performing background screening on the training samples in advance [54], [55]. All these strategies can further increase the reconstruction error for anomalies. Very recently, Li et al. [56] integrated spatial and spectral characteristics via a two-stream convolutional AE architecture, with a parameter-free clustering approach chosen to select background samples.

D. Motivation and Contributions

Among the DL-based HAD methods described above, similarity measurement models tend to involve a complicated training sample generation process, and the model's generalizability is relatively limited. Meanwhile, most reconstruction models take the input HSI as the learned target, which is prone

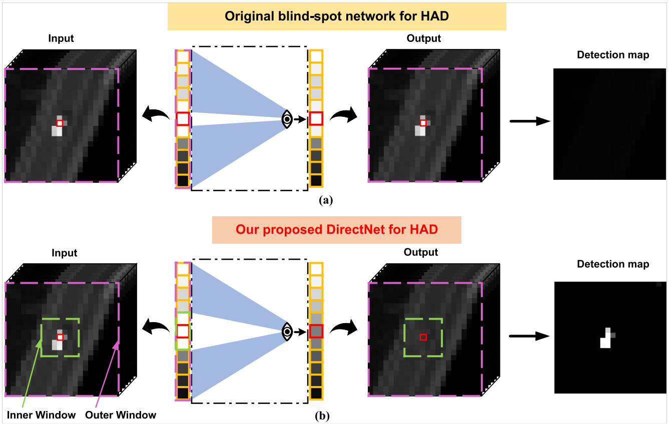


Fig. 1. Blind-spot reconstruction network has a blind area in the receptive field corresponding to a single central pixel (i.e., the white area invisible to the network/eye). Therefore, the network has to predict/reconstruct the central pixel (represented by a red rectangle) using only the surrounding pixels (the blue area seen by the eye). (a) Original blind-spot network contains a blind spot in the center of the receptive field. Here, the reconstruction of anomalous pixels is interfered with by neighboring pixels that are also anomalous, ultimately leading to undesirable results in relatively large-sized targets. (b) Our proposed DirectNet improves the original blind-spot network by conceiving a dual-window blind-block reconstruction model. Specifically, the original receptive field of DirectNet is divided into a dual window, where the blind-block area is considered the inner window (represented by a green dashed rectangle) and the receptive field outside the inner window (represented by a magenta dashed rectangle) is regarded as the outer window. Fortunately, DirectNet possesses promising detection performance for both large- and small-sized targets, thus extending the application range of blind-spot models.

to lead to the identity mapping being learned [57], [58], i.e., parts of the anomalies will inevitably appear in the reconstructed HSI, although it is expected that the model serves as a pure background reconstructor. Therefore, the obtained results manifested by the reconstruction errors make it difficult to separate anomalies from the background, resulting in a degradation of the model performance.

To address the aforementioned issues, we opted to perform HAD using a self-supervised learning technique. Self-supervised learning is similar to unsupervised learning in that only unlabeled HSIs are required, yet it generates supervision signals from the raw data for network training [59], [60]. Nowadays, self-supervised learning has proven to be effective in diverse topics related to HSI processing, such as denoising [61], fusion [62], [63], spectral unmixing [64], and classification [65]. More recently, the blind-spot network [66] (as a representative of self-supervised learning) has been introduced to HAD for accurately identifying small-sized subpixel anomalous targets in HSIs [67]. However, this technique is unable to detect large-sized targets effectively in a direct way. To explain the challenge of detecting large-sized targets using the original blind-spot network, a toy example is given in Fig. 1. We emphasize that the original blind-spot network possesses a special receptive field centered on a blind spot, implying that the network fails to see the spectral information of the receptive field's central pixel. Hence, the network's prediction value for the blind spot depends only on its neighboring pixels, excluding its own information and avoiding identity mapping [68]. When the central pixel is an anomaly (i.e., significantly different from its surrounding pixels), the blind-spot network will not predict/reconstruct the central one adequately, leading to an increased reconstruction error and a high anomaly score. Accordingly, the architecture of blind-spot networks is well-suited to perform the HAD task. Nevertheless, as it can be seen in Fig. 1(a), the original blind-spot network does not succeed in recognizing large-sized anomaly targets. This is because the network is influenced by the neighboring pixels containing the target's spectral information when predicting the central anomalous pixel, which causes large-sized anomalies to exhibit low reconstruction errors and low anomaly scores.

To tackle this challenge, we conceive a sliding dual-window-inspired reconstruction network for HAD in a self-supervised manner. As shown in Fig. 1(b), our network divides the original receptive field into a dual window, where only the outer window pixels are exploited to predict the central pixel, while the inner window works as a guard window against potential contaminating effects coming from anomalous pixels. When the central anomalous sample is reconstructed, sparsely distributed and small-occupancy targets with a certain spatial size tend to lie in the inner window, whereas the network will be unable to see the inner window information, so that no anomalous properties are learned. Those backgrounds with large scales and homogeneous properties always remain located in the outer window, which impels the samples predicted by the network to converge to the surrounding background distribution, resulting in an increase in the reconstruction error of anomalies. Specifically, first, training patch pairs are constructed from the unlabeled part of the observed HSI. Then, the designed background reconstruction model cannot see the center block of the receptive field (inner window) and has to predict/reconstruct the central pixel using only the receptive field area outside the center block (outer window), which is in accordance with the idea of the dual-window model, weakening the ability to fit anomalous features. Finally, the trained network turns out to be a superb background reconstructor, and the reconstruction error is employed to reflect the degree of anomaly of each pixel. Compared with existing HAD methods, the core contributions of our proposed sliding **D**ual-window-**i**nspired **rec**ons**t**ruction **Net**work (DirectNet) are summarized as threefold.

1) We propose a novel blind-block architecture-based network for HAD purposes, which operates in a fully self-supervised learning manner, i.e., generating training pairs from an unlabeled observed HSI, leading to impressive detection performance (surpassing that achieved by unsupervised reconstruction models).
2) To fully leverage the spatial–spectral features of an HSI, we propose an adaptive depth reconstruction network. This network adjusts its depth according to the size of the input image patch, ensuring that the network's receptive field aligns with the dimensions of the input patch.
3) To address the limitations of blind-spot networks in handling large-size targets, we propose a novel dual-window-inspired blind-block network. Unlike traditional blind-spot networks that focus on one-pixel blind spots,

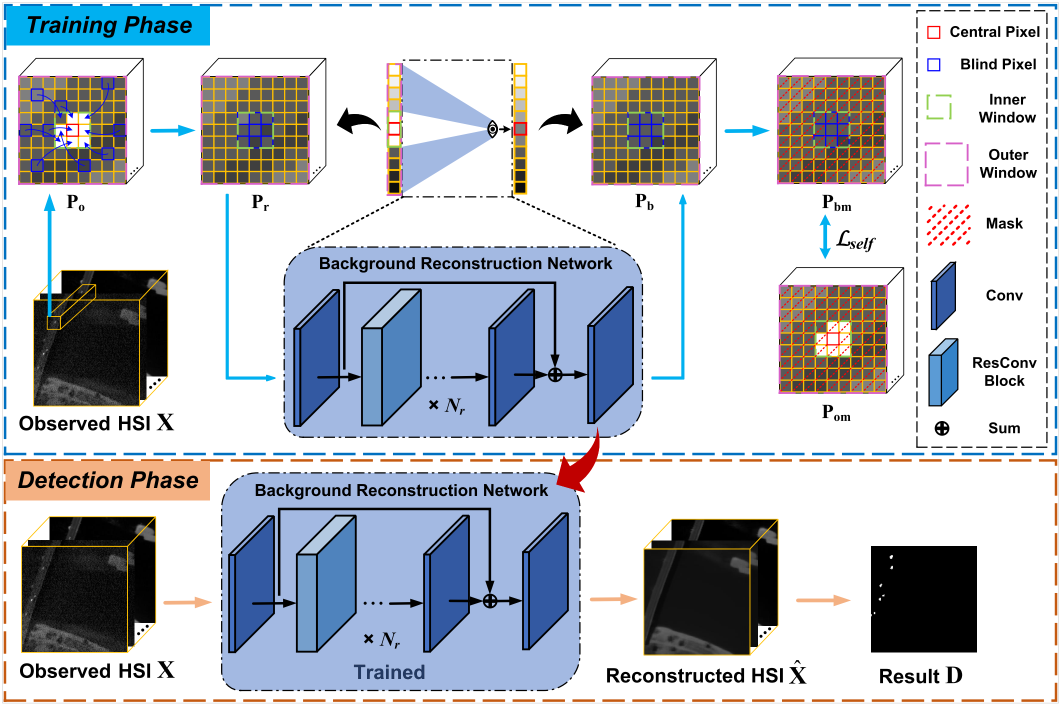


Fig. 2. Flowchart of the proposed DirectNet.

our approach considers a large-size block as the blind area of the receptive field. We erase the center block information of the image patch, meaning that the network is unable to see this information when reconstructing the central pixel. The blind center block serves as an inner window, which prevents large-sized anomalies located in the inner window from influencing the reconstruction of the central pixel. Simultaneously, the regions of the receptive field outside the center block are considered the outer window. By leveraging the surrounding outer window pixels, our network predicts/reconstructs the central pixel. This design forces the network to rely on the outer window information, leading to significant reconstruction errors for anomalous targets. Consequently, our DirectNet shows promising potential in recognizing both large- and small-sized targets, thus expanding the application scope of original blind-spot networks.

The rest of this article is organized as follows. We give the details of the proposed DirectNet in Section II. In Section III, comprehensive experimental results between DirectNet and state-of-the-art detectors are discussed. Finally, the conclusion is summarized in Section V.

II. Proposed Anomaly Detector: DirectNet

We designed a background reconstruction network inspired by a sliding dual-window model, called DirectNet, whose overall flowchart is shown in Fig. 2. More specifically, DirectNet can be divided into two stages.

1) *Training Phase:* First, pairs of input and pseudo-labeled data (in the form of patches) are obtained from the observed HSI, and then, they are fed into the background reconstruction network. Finally, a masking scheme is devised to modify the self-supervised objective function. (The mask here is different from the mainstream definition, as described in [69]. Instead of representing the occluded regions during training, it refers to the portions that are not utilized in the calculation of the loss function.) The above process forces the network not to see the information of the center block of the receptive field (inner window), but instead to reconstruct the central pixel using the remaining receptive field (outer window).
2) *Detection Phase:* The whole observed HSI is fed into the well-trained DirectNet to estimate the expected pure background HSI, and the reconstruction error is treated as the final detection result. Next, we fully detail each part of DirectNet.

A. Problem Formulation

Let us consider an HSI as $\mathbf{X} = \{\mathbf{x}_i \in \mathbb{R}^{L\times 1}\}_{i=1}^{R\times C}$, where R, C, and L, respectively, represent the row number, column number, and band/channel number of $\mathbf{X}$, and $\mathbf{x}_i$ denotes the ith spectral vector pixel of size L. The proposed DirectNet is expected to be a superior background reconstructor using a self-supervised strategy, i.e., only using the observed HSI $\mathbf{X}$ without any labeled background and anomaly information, to well-identify the anomalous targets. Furthermore, DirectNet divides its own receptive field into a dual window, which is illustrated in Fig. 3. The pixel information lying in the inner window is invisible to the network when reconstructing the central pixel. Our DirectNet empowers the feature expression of the background, generating a small reconstruction error for backgrounds and a large reconstruction error for anomalies.

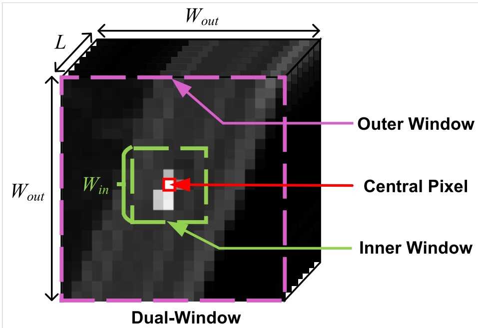


Fig. 3. Illustration of the dual window provided by our DirectNet. Note that only the outer window pixels are employed to predict the central pixel.

Therefore, the reconstruction error showcases the degree of anomaly for each pixel naturally, facilitating the recognition of anomalous targets in HSIs.

B. Training Phase

Here, we introduce the training phase of DirectNet, i.e., the generation of training patch pairs as well as the specific structure.

Let $\mathbf{p_{o}}_i \in \mathbb{R}^{W_{\text{out}} \times W_{\text{out}} \times L}$ denote a patch centered at pixel $\mathbf{x}_i$, where W_{out} is the width of the patch (usually an odd number). Symbol $\mathbf{P_o} = \{\mathbf{p_{o}}_i\}_{i=1}^{R\times C}$ denotes the patches extracted from the HSI. In order to keep the network from seeing the pixels in the center block (denoted as $\mathbf{p_{c}}_i \in \mathbb{R}^{W_{\text{in}} \times W_{\text{in}} \times L}$) of $\mathbf{p_{o}}_i$, we erase the spectral information covering the corresponding center block. It is implemented by randomly picking $W_{\text{in}} \times W_{\text{in}}$ pixels from the region excluding $\mathbf{p_{c}}_i$ to replace all pixels of $\mathbf{p_{c}}_i$, with the purpose of forming $\mathbf{P_r} = \{\mathbf{p_{r}}_i\}_{i=1}^{R\times C}$, where $\mathbf{p_{r}}_i \in \mathbb{R}^{W_{\text{out}} \times W_{\text{out}} \times L}$ is the ith patch that removes the information of $\mathbf{p_{c}}_i$. Indeed, $\mathbf{P_r}$ and $\mathbf{P_o}$ are considered as the input data and learned target of DirectNet. A tailored masking strategy will be subsequently devised to allow the network to concentrate its efforts on reconstructing the central pixel of each patch. The above procedure creates an inner window (guard window) of size $W_{\text{in}} \times W_{\text{in}}$ for the corresponding receptive field of the network during the prediction of the central pixel, enabling the need to avoid the pixels located in the inner window whenever the central pixel is reconstructed. In the case of reconstructing anomalous pixels, targets of a certain spatial size normally fall in the inner window, which lessens the contribution of anomalous information. This is unlike the original blind-spot network, as the setting of the inner window endows our DirectNet with the ability to detect targets of different sizes.

Next, a ResNet block-based background reconstruction network (whose receptive field is adaptively matched to the training patch size) is designed. The number of network layers, i.e., the number of adopted ResNet blocks, is automatically derived from the patch size, avoiding manually configured parameters. Since the size of the network's receptive field and patch aligned well, this ensures that the reconstruction of the central pixel will exploit all the spatial–spectral information of the patch. Nevertheless, owing to the setting of the inner window (meaning that the network is unable to see this area), the pixels actually devoted to the reconstruction come from only the contents of the patch lying outside the inner window. Clearly, the receptive field of the network for reconstructing the central pixel is partitioned into a dual window, where the inner window is of size $W_{\text{in}} \times W_{\text{in}}$ and the outer window is of size $W_{\text{out}} \times W_{\text{out}}$ (with the same spatial size as the training patch), and when the network reconstructs the central pixel, it only utilizes the surrounding pixels in the outer window. Concretely, the detailed network structure is presented in Fig. 4, where the convolution kernel size of all convolution layers is 3 × 3. The first convolution layer is followed by the Rectified Linear Unit (ReLU) function, and the penultimate convolution layer is followed by a batch normalization layer. We applied N_r ResNet blocks, where $N_r = (W_{\text{out}} - 7)/4$ that is derived from the network's receptive field. For example, when the outer window size is 15×15 (i.e., W_{out} is 15), our DirectNet contains two ResNet blocks. In the ResNet block, there are two convolution layers both followed by the batch normalization layer, and the first layer also takes the ReLU function. The pixel-level summation (the same as given next) is performed on the output and input feature maps to realize skip connections. Meanwhile, to fuse the beginning low-level visual with high-level semantic features, the feature maps obtained from the first and penultimate convolution layers are summed and fed into the last convolution layer. We denote the proposed background reconstruction model as $\mathcal{F}_r$, parameterized by θ. Then, $\mathbf{P_r}$ is input into $\mathcal{F}_r$ to get the output, i.e.,

$$\mathbf{P_b} = \mathcal{F}_r(\mathbf{P_r}; \theta) \tag{1}$$

where $\mathbf{P_b} = \{\mathbf{p_{b}}_i\}_{i=1}^{R\times C}$ and $\mathbf{p_{b}}_i \in \mathbb{R}^{W_{\text{out}} \times W_{\text{out}} \times L}$ indicates the patch corresponding to $\mathbf{p_{r}}_i$, in which L is the number of convolution channels in the last layer, which is equal to the number of bands of $\mathbf{X}$.

Given that $\mathbf{P_o}$ is the supervision information for $\mathbf{P_r}$, the optimization goal of the proposed DirectNet is to make $\mathbf{P_b}$ (excluding the center block information) converge to $\mathbf{P_o}$. In particular, DirectNet only exploits the contents within the outer window of $\mathbf{P_o}$ to reconstruct the corresponding central pixel, which inspires us to impose a mask-learning scheme to enable $\mathcal{F}_r$ to specialize in reconstructing/predicting the central pixel of the patch, serving the dual-window model construction. More specifically, we additionally applied a binary mask to $\mathbf{P_b}$ and $\mathbf{P_o}$ to obtain $\mathbf{P_{bm}}$ and $\mathbf{P_{om}}$ (see Fig. 2), aiming to only calculate the loss at the central pixel position during backpropagation. By now, only the outer window pixels of the receptive field are available for DirectNet to reconstruct the central pixel, in which the homogeneous background would commonly be located inside the outer window, impelling the background samples to be reconstructed successfully. Meanwhile, it induces that each pixel reconstructed by DirectNet will satisfy the background distribution of the outer window, weakening the model's power to fit anomalous targets, and motivating its promise to be a well-performing background reconstructor.

Our DirectNet $\mathcal{F}_r(\cdot; \theta)$ is trained using an ℓ_1 loss-based objective function as follows:

$$\begin{aligned}\hat{\theta} &= \arg\min_{\theta} \mathcal{L}_{self} \\ &= \arg\min_{\theta} \|\mathcal{F}_r(\mathbf{P_r}; \theta) \odot \mathbf{M} - \mathbf{P_o} \odot \mathbf{M}\|_1\end{aligned}$$

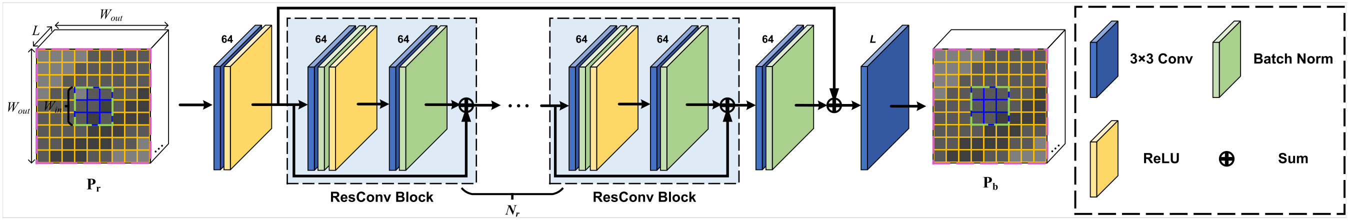


Fig. 4. Details of the proposed overall network structure (the numbers above Conv, such as 64, represent the number of output channels).

$$
\begin{aligned}
&= \arg\min_{\theta} \|\mathbf{P_b} \odot \mathbf{M} - \mathbf{P_o} \odot \mathbf{M}\|_1 \\
&= \arg\min_{\theta} \|\mathbf{P_{bm}} - \mathbf{P_{om}}\|_1
\end{aligned} \tag{2}
$$

where $\mathbf{M} = \{\mathbf{m}_i\}_{i=1}^{R\times C}$ represents the patches of binary mask, $\mathbf{m_i} \in \{0,1\}^{W_{\text{out}}\times W_{\text{out}}\times L}$ is the mask corresponding to $\mathbf{p_{b\mathit{i}}}$ and $\mathbf{p_{o\mathit{i}}}$ (see Fig. 2), the position of the one in $\mathbf{m}_i$ denotes the position of the central pixel, and the rest of the elements are all zeros. Symbol $\odot$ denotes elementwise multiplication. Here, for better generalization, we adopt the ℓ_1 loss instead of the typical ℓ_2 loss that is more attentive to larger errors but less sensitive to smaller ones [70]. This ℓ_1 norm loss function is based on the mean absolute error, which requires calculating the absolute value of the difference between the output predicted data and the supervision information, and its gradient descent process is based on the adaptive moment estimation (Adam) stochastic optimization algorithm [71] that comes with the PyTorch framework.

C. Detection Phase

In the detection phase, the observed HSI $\mathbf{X}$ is fed into the well-trained DirectNet, parameterized by $\hat{\theta}$, to acquire the final reconstructed image (denoted as $\widehat{\mathbf{X}} = \{\hat{\mathbf{x}}_i \in \mathbb{R}^{L\times 1}\}_{i=1}^{R\times C}$), i.e.,

$$
\widehat{\mathbf{X}} = \mathcal{F}_r(\mathbf{X}; \hat{\theta}). \tag{3}
$$

After the aforementioned self-supervised training procedure, the proposed DirectNet behaves as an efficient background reconstruction model; thus, the output $\widehat{\mathbf{X}}$ becomes a pure background image, and naturally, anomalies have larger reconstruction errors compared to backgrounds. Finally, the reconstruction error is recognized as a detection result as follows:

$$
D_i = \left\|\mathbf{x}_i - \hat{\mathbf{x}}_i\right\|_2 \tag{4}
$$

where $\mathbf{x}_i$ and $\hat{\mathbf{x}}_i$ denote the ith pixels of $\mathbf{X}$ and $\widehat{\mathbf{X}}$, respectively, and D_i stands for the abnormal response value of the ith pixel and constitutes the final HAD map $\mathbf{D} = \{D_i\}_{i=1}^{R\times C}$. This ℓ_2 norm reconstruction error is based on the mean square error, which requires calculating the square value of the difference between the output predicted data and the original one. The key steps of the proposed DirectNet are summarized in Algorithm 1.

Algorithm 1 DirectNet

Input: Observed HSI $\mathbf{X} = \{\mathbf{x}_i \in \mathbb{R}^{L\times 1}\}_{i=1}^{R\times C}$;
Parameters: 1) width of inner window W_{in}, 2) width of outer window W_{out}.
Output: Final detection result $\mathbf{D} = \{D_i\}_{i=1}^{R\times C}$.

Stage 1: Training Phase;
for $i = 1 : N(N = R\times C)$ **do**
  Construct patch $\mathbf{p_{o\mathit{i}}} \in \mathbb{R}^{W_{\text{out}}\times W_{\text{out}}\times L}$ centered at pixel $\mathbf{x}_i$;
  Replace all pixels in the center block $\mathbf{p_{c\mathit{i}}} \in \mathbb{R}^{W_{\text{in}}\times W_{\text{in}}\times L}$ of $\mathbf{p_{o\mathit{i}}}$ with $W_{\text{in}} \times W_{\text{in}}$ pixels randomly selected from the region outside of $\mathbf{p_{c\mathit{i}}}$;
  Get the patch $\mathbf{p_{r\mathit{i}}} \in \mathbb{R}^{W_{\text{out}}\times W_{\text{out}}\times L}$ that removes the information of $\mathbf{p_{c\mathit{i}}}$;
end
Obtain pairs of $\mathbf{P_r} = \{\mathbf{p_{r\mathit{i}}}\}_{i=1}^{R\times C}$ and $\mathbf{P_o} = \{\mathbf{p_{o\mathit{i}}}\}_{i=1}^{R\times C}$ as input data and learned target for network training, respectively;
for $i = 1 : N$ **do**
  Train a reconstruction network $\mathcal{F}_r$ with $N_r = (W_{\text{out}} - 7)/4$ ResNet blocks by minimizing loss function Eq. (2);
end
Obtain a well-trained network $\mathcal{F}_r$, parameterized by $\hat{\theta}$;
Stage 2: Detection phase;
Acquire a reconstructed HSI by feeding $\mathbf{X}$ into the well-trained DirectNet, i.e., $\widehat{\mathbf{X}} = \mathcal{F}_r(\mathbf{X}; \hat{\theta})$;
Calculate the degree of anomaly D_i of each pixel in $\mathbf{X}$ by Eq. (4);

III. Experimental Results

In this section, the superior performance of the proposed DirectNet is validated qualitatively and quantitatively through extensive experiments.

A. Datasets

1) Salinas Dataset: This is a synthetic image collected by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) covering the Salinas valley [72], and we adopted the anomaly target implantation approach of [73], which follows a linear mixing model [74], [75], [76], i.e., $\mathbf{n} = f_a\mathbf{a} + (1 - f_a)\mathbf{b}$, where $\mathbf{a}, \mathbf{b}, \mathbf{n} \in \mathbb{R}^{L\times 1}$ are spectral vectors of the anomaly, background, and final mixture of the two, respectively, and f_a denotes the nonnegative abundance fraction of $\mathbf{a}$. The 16 anomalous objects (buildings) with a spatial size of 3×3 were randomly embedded into this sub-HSI, and the abundance fraction of $\mathbf{a}$ for these targets begins at 0.05 and increases by 0.04 in turn.

2) Pavia Dataset: It was captured by the Reflective Optics System Spectrographic Imaging System (ROSIS-03) sensor

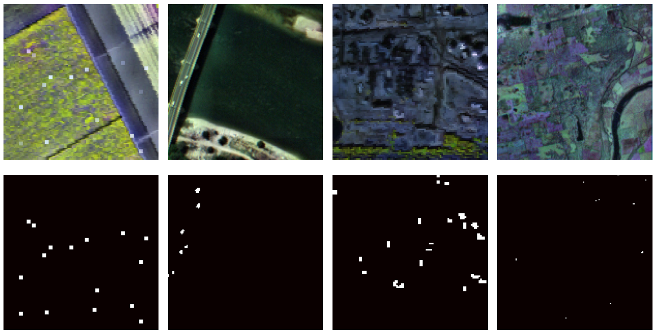


Fig. 5. Pseudocolor images and ground truth maps of the four considered HSI datasets. (a) Salinas. (b) Pavia. (c) El Segundo. (d) Indiana.

TABLE I
DETAILS OF THE EXPERIMENTAL DATASETS

Data set	Location	Sensor	Image size	Resolution
Salinas	Salinas	AVIRIS	$120 \times 120 \times 204$	3.7m
Pavia	Pavia	ROSIS-03	$150 \times 150 \times 102$	1.3m
El Segundo	El Segundo	AVIRIS	$100 \times 100 \times 224$	7.1m
Indiana	Indiana	Hyperion	$150 \times 150 \times 155$	10m

over the city of Pavia [77]. There are several vehicles embedded in the bridge as anomalous targets.

3) El Segundo Dataset: It was collected by AVIRIS and covered an urban scene in El Segundo city [78], in which the anomalies mainly consist of storage tanks on the oil refinery.

4) Indiana Dataset: It was acquired from the Earth Observing One (EO-1) Hyperion sensor over the Indiana area [79], and some storage bins and roofs of small size are considered to be anomalies in this image.

Notably, the above four images contain both small- and large-sized anomalous targets, and more detailed features of these images are shown in Table I. In addition, the pseudocolor images and ground truth maps of targets are shown in Fig. 5.

B. Comparison Algorithms and Evaluation Metrics

1) Comparison Algorithms: To assess the efficacy of the proposed DirectNet, some well-established competitive methods selected for comparison embrace four statistical modeling-based methods (i.e., GRX [9], FrFE [13],[1] LRX [10], and 2S-GLRT [15][2]), two representation learning-based methods (i.e., CRD [30] and LSDM-MoG [35][3]), and four DL-based methods (i.e., GAED [48],[4] RGAE [49],[5] Auto-AD [50],[6] and BS^{3}LNet [67]). Among the above detectors, all based on cutting-edge DL technology, GAED, RGAE, Auto-AD, and BS^{3}LNet methods are designed to enhance the capacity of models to reconstruct backgrounds, and all adopt the residual spectral error between the original and reconstructed HSIs to measure the degree of anomaly of each pixel. This remains in line with the motivation of our proposed DirectNet. Meanwhile, the blind-spot network architecture endows BS^{3}LNet with the power to detect subpixel small-sized anomalies in the HSI. Hence, the differences between the detection effect of these five methods require to be involved in emphasizing the effectiveness of coupled dual window with deep reconstruction networks. For the proposed DirectNet, it is implemented by PyTorch on a GeForce RTX 2080 Ti GPU with 11-GB memory. The batch size and the learning rate were set to 100 and 1e^{-4}, respectively, for all datasets. The spatial sizes of the input patches for the four datasets (namely, Salinas, Pavia, El Segundo, and Indiana) were set to 15×15, 19×19, 19×19, and 23×23, with the number of training epochs for the four datasets set to 700, 1000, 700, and 950, respectively.

2) Evaluation Metrics: To evaluate the performance of the different popular compared detectors, some well-known criteria are adopted, i.e., receiver operating characteristic (ROC), the area under the ROC curve (AUC) [80], and the statistical separability map. Among them, the ROC curve reveals the relationship between the probability of detection and the false alarm rate (FAR), revealing the detection accuracy of the model in a holistic manner. The corresponding AUC value can assess the detection accuracy quantitatively. An excellent detector would carry an ROC curve in near proximity to the upper left corner, with an AUC value near 1. Moreover, the separability map provides the statistical range of the abnormal response values at the pixel location of the background and anomalous classes, and these values are distributed in two separated boxes (blue for the background and red for the anomaly), each with a statistical interval from 10% to 90%. Clearly, if the detector demonstrates a predominant ability to extrude anomalies and suppress backgrounds, the blue box will be at the exact bottom and narrow width, with a big interval from the red box.

C. Detection Performance for Different Methods

Figs. 6–9 visualize the HAD colored results obtained by the 11 considered methods on the four considered datasets, where the first one in each figure is the ground truth of detected anomalous targets, and the rest are the resulting maps for different detectors. Also, Figs. 10 and 11 show the ROC curves and separability boxplots of these 11 methods, whose AUC scores are listed in Table II. Note that the best results are marked in boldface (the same as given next).

Across the board, Figs. 6–9 show that the proposed DirectNet attains an acceptable tradeoff in anomaly detectability and background suppressibility, and its corresponding detection maps are closest to the ground truth. Considering the Salinas dataset as an example, GRX and FrFE do not work satisfactorily because the shapes of 16 targets are barely visible in the result maps. Although LSDM-MoG, GAED, and RGAE perform relatively well in highlighting anomalies, they maintain some background structural information and so suffer from varying degrees of false alarms. LRX, 2S-GLRT, CRD, Auto-AD, and BS^{3}LNet can restrain the background to

[1]https://github.com/xudongzhao461/Hyperspectral-Anomaly-Detection-by-Fractional-Fourier-Entropy

[2]https://github.com/zephyrhours/Hyperspectral-Anomaly-Detection-2S-GLRT

[3]https://github.com/l7170/LSDM-MoG

[4]https://github.com/pei-xiang/GAED

[5]https://github.com/FGH00292/Hyperspectral-anomaly-detection-with-RGAE

[6]https://github.com/RSIDEA-WHU2020/Auto-AD

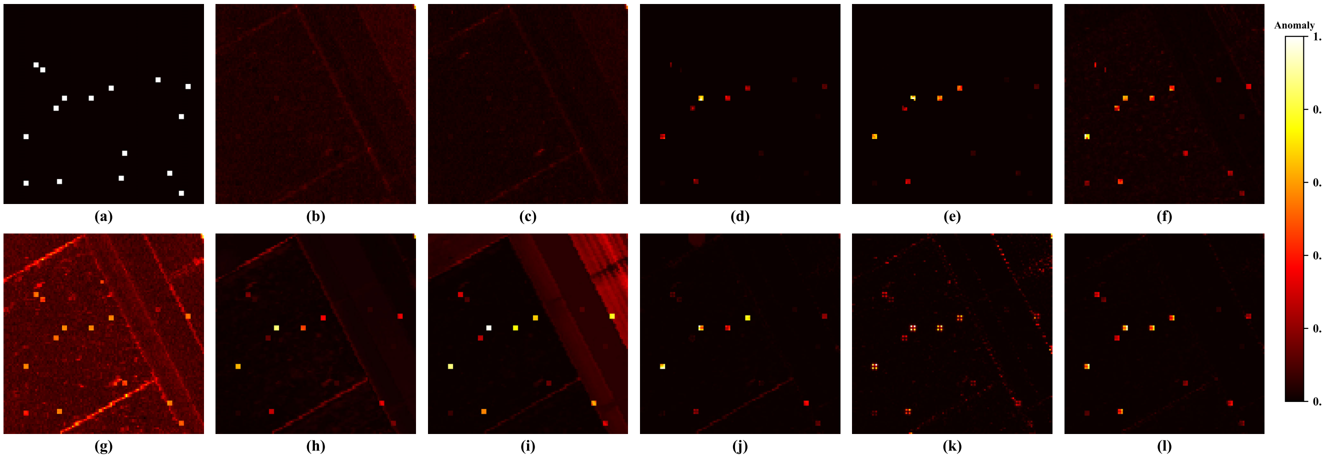


Fig. 6. HAD colored maps obtained by different algorithms for the Salinas dataset. (a) Ground truth. (b) GRX. (c) FrFE. (d) LRX. (e) 2S-GLRT. (f) CRD. (g) LSDM-MoG. (h) GAED. (i) RGAE. (j) Auto-AD. (k) BS^{3}LNet. (l) DirectNet.

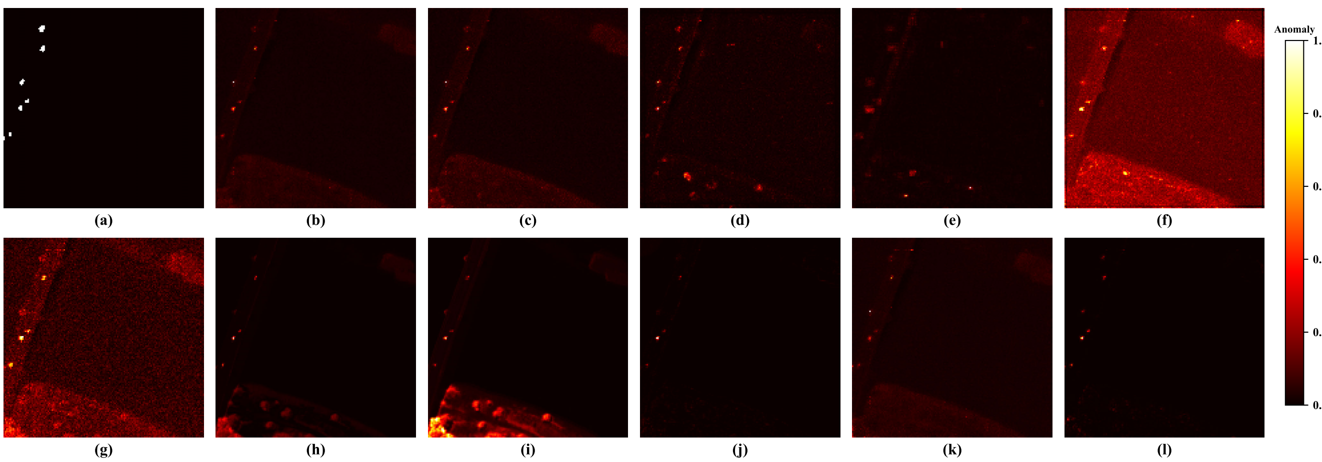


Fig. 7. HAD colored maps obtained by different algorithms for the Pavia dataset. (a) Ground truth. (b) GRX. (c) FrFE. (d) LRX. (e) 2S-GLRT. (f) CRD. (g) LSDM-MoG. (h) GAED. (i) RGAE. (j) Auto-AD. (k) BS^{3}LNet. (l) DirectNet.

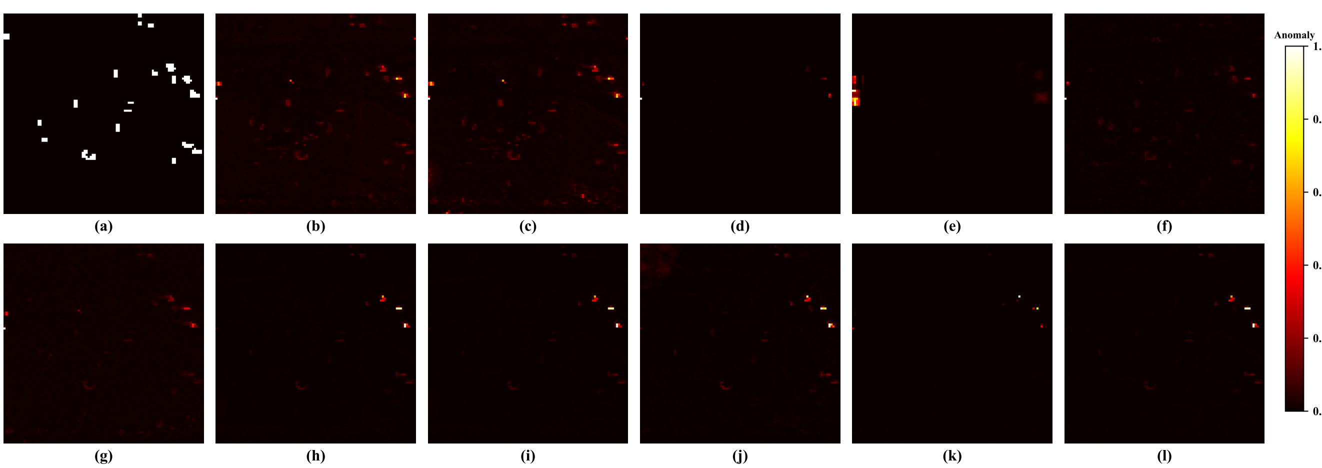


Fig. 8. HAD colored maps obtained by different algorithms for the El Segundo dataset. (a) Ground truth. (b) GRX. (c) FrFE. (d) LRX. (e) 2S-GLRT. (f) CRD. (g) LSDM-MoG. (h) GAED. (i) RGAE. (j) Auto-AD. (k) BS^{3}LNet. (l) DirectNet.

a certain extent but cannot effectively highlight the anomalous targets. Conversely, our DirectNet avoids the aforementioned issues and can accurately identify the embedded pixels with high abnormal response values and low false alarms.

Another example is the Pavia dataset. CRD, LSDM-MoG, and RGAE can highlight these anomalous targets, yet they outline the boundaries of backgrounds, making the distinction between backgrounds and anomalies not clear. Although GRX, FrFE, LRX, 2S-GLRT, GAED, and BS^{3}LNet can locate most of the vehicles, they risk misjudging the background. Auto-AD and DirectNet can suppress background interference to a large degree, yet our DirectNet is more capable of enhancing

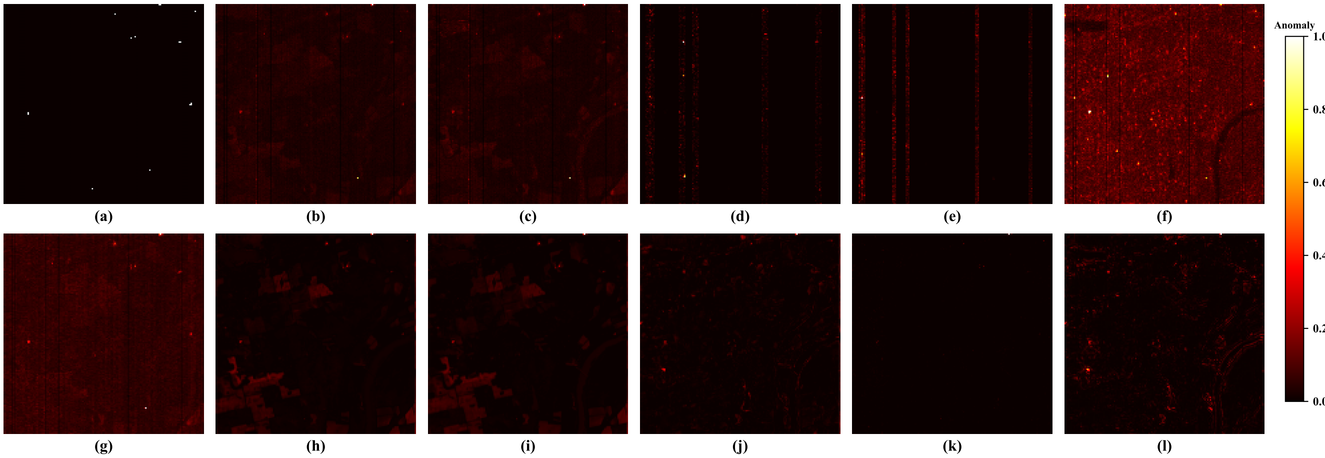


Fig. 9. HAD colored maps obtained by different algorithms for the Indiana dataset. (a) Ground truth. (b) GRX. (c) FrFE. (d) LRX. (e) 2S-GLRT. (f) CRD. (g) LSDM-MoG. (h) GAED. (i) RGAE. (j) Auto-AD. (k) BS^{3}LNet. (l) DirectNet.

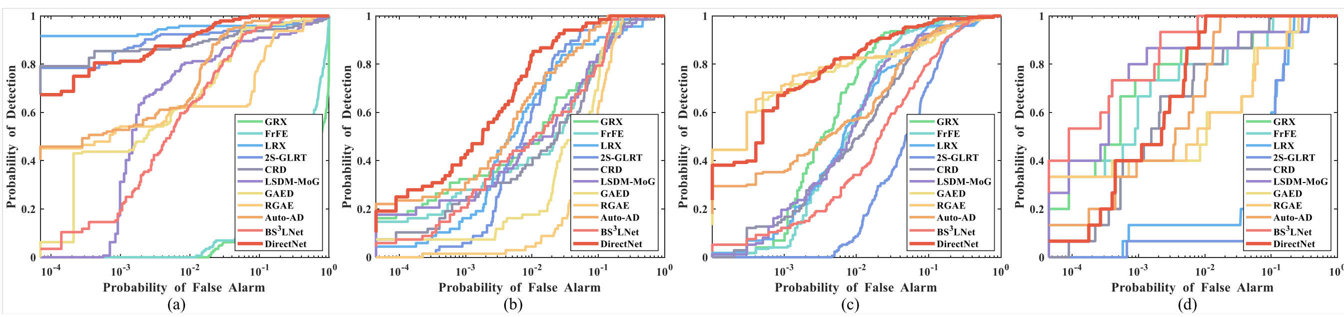


Fig. 10. ROC curves obtained by different detectors for the four considered datasets. (a) Salinas. (b) Pavia. (c) El Segundo. (d) Indiana.

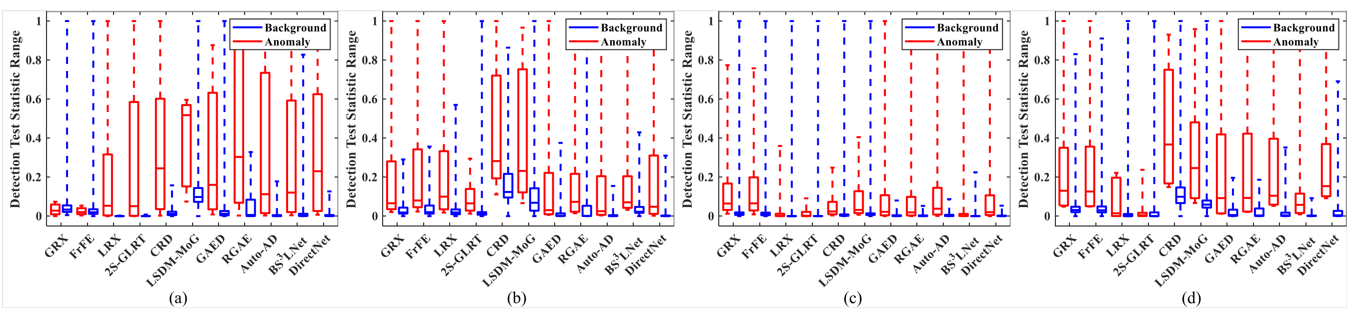


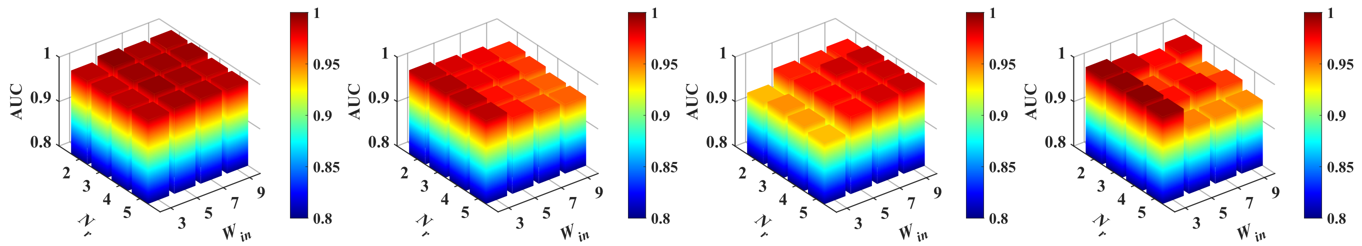
Fig. 11. Separability maps obtained by different detectors on the four considered datasets. (a) Salinas. (b) Pavia. (c) El Segundo. (d) Indiana.

anomalous characteristics compared to Auto-AD while yielding a lower FAR. Analogous results are displayed on the other two datasets, and the above visualizations show that DirectNet holds a better capacity for separating targets from backgrounds. The main reason lies in the fact that the proposed DirectNet couples the sliding dual-window model in a self-supervised way, indicating promising performance in strengthening anomalous characteristics and restraining background interference.

In addition, other metrics were operated to evaluate the detection performance of different algorithms from qualitative and quantitative views. In Fig. 10, overall, the red curve of DirectNet lies mostly above the other colored curves and generally exceeds the other detectors. Unsurprisingly, compared with the other ten competing methods, DirectNet achieves the highest AUC values on the first three datasets and second place on the Indiana dataset. BS^{3}LNet, known for its effectiveness in detecting small-sized subpixel anomalous targets while excelling on the Indiana dataset, performs poorly on the other three datasets with large-sized targets. To further verify the strong background suppressibility and anomalous target separability of DirectNet, Fig. 11 draws separability boxplots of all detectors regarding backgrounds and anomalies. On these four datasets, the red anomaly boxes of DirectNet are normally not located at the highest place, but its blue background boxes are at a rock-bottom level and narrow enough, implying that DirectNet is well qualified for restraining the background. At the same time, on the Salinas and Indiana datasets, DirectNet can ensure a larger gap between the two boxes and swimmingly extract anomalies from the suppressed backgrounds. Apparently, this accords with the visual analysis effects described earlier.

TABLE II
AUC VALUES OF THE 11 CONSIDERED DETECTORS FOR DIFFERENT DATASETS

Data Set	Detectors and AUC values										
	GRX	FrFE	LRX	2S-GLRT	CRD	LSDM-MoG	GAED	RGAE	Auto-AD	BS^{3}LNet	DirectNet
Salinas	0.3798	0.5270	0.9747	0.9779	0.9642	0.9530	0.9736	0.9365	0.9887	0.9783	**0.9971**
Pavia	0.9538	0.9498	0.9590	0.9802	0.9407	0.9448	0.9394	0.9094	0.9820	0.9571	**0.9904**
El Segundo	0.9855	0.9794	0.9640	0.9151	0.9637	0.9699	0.9682	0.9694	0.9660	0.9395	**0.9863**
Indiana	0.9896	0.9850	0.9018	0.8676	0.9877	0.9782	0.9535	0.9602	0.9943	**0.9990**	0.9971
Average	0.8272	0.8603	0.9499	0.9352	0.9641	0.9615	0.9587	0.9439	0.9828	0.9685	**0.9927**



(a) (b) (c) (d)

Fig. 12. AUC values of the proposed DirectNet with inner windows of different widths (W_{in}) and ResNet blocks of different numbers (N_r) on the four considered datasets. (a) Salinas. (b) Pavia. (c) El Segundo. (d) Indiana.

All told that, from the findings by visual inspection, the proposed DirectNet presents remarkable visual results. Besides, the comparison and analysis of the ROC curves, AUC scores, and separability boxplots for different methods reveal that DirectNet achieves the highest average AUC of 0.9927, which is 0.0099 larger than that of the second-best Auto-AD, as reported in Table II. In addition, the FAR of DirectNet remains generally at a low value, further achieving a satisfactory balance between anomaly detectability and background suppressibility, thus affirming the efficacy of DirectNet.

D. Parameter Analysis

The proposed DirectNet involves two parameters: the width of the inner window W_{in} and the number of ResNet blocks N_r (whose setting is adaptive to the width of the outer window, i.e., $W_{\text{out}} = 4 \times N_r + 7$). It is noteworthy that the feasible value of W_{in} needs to be an odd number greater than 1. If W_{in} is equal to 1, then DirectNet degenerates into a blind-spot network with a coupled single-window model. Among them, the values of W_{in} and N_r for all the datasets are set to {3, 5, 7, 9} and {2, 3, 4, 5} (i.e., W_{out} is set to {15, 19, 23, 27}), respectively. Then, the AUC scores under different parameter settings are summarized in Fig. 12.

It can be found from Fig. 12 that the detection accuracy of DirectNet (coupled with dual-window of different sizes) does not exhibit significant fluctuations on the Salinas dataset. For the El Segundo dataset (with large-sized targets), W_{in} should be chosen as a larger value, and for datasets with small-sized targets (such as Pavia and Indiana), the smaller value should be selected for W_{in} to further promote the performance of DirectNet. As long as the inner window size is selected appropriately, the outer window size (number of ResNet blocks) brings little impact, manifesting the stability of DirectNet in the task of detecting anomalous targets of diverse sizes. For the four datasets containing targets of different spatial sizes, the parameter W_{in} is fixed to 5, 3, 9, and 3 as the optimal setting.

E. Ablation Study

This section evaluates the contribution of each component to the proposed DirectNet on all datasets. As a baseline, we employ a traditional unsupervised learning-based reconstruction model (denoted as USLNet), which uses the same framework as DirectNet, except for the fact that it trains the background reconstruction network directly with the original patches, without removing any information. By placing the inner window in the center of the receptive field (meaning that this information is not visible to the network) and the outer window (corresponding to the receptive field area) outside of the center block, our DirectNet couples a dual-window model, which is well-suited for detecting anomalous targets of different sizes. When there is no inner window (i.e., the width of the inner window is 1), DirectNet degenerates into a blind-spot network with a single-window model (denoted as BaseNet). Meanwhile, we also compared the effect of different types (i.e., ℓ_1 and ℓ_2) of reconstruction errors on the detection performance. The details of AUC comparisons on these datasets are shown in Table III.

As seen from Table III, BaseNet has some advantages over the unsupervised learning-based method on most datasets, especially for the Indiana dataset containing slightly small-sized targets, where BaseNet achieves an overwhelming superiority by virtue of its blind-spot network architecture. However, when reconstructing the pixels within large-sized anomalous targets, both the unsupervised learning-based USLNet and the self-supervised learning-based BaseNet (joint spatial–spectral learning-based models) are prone to interference from neighboring anomalous pixels, which causes

TABLE III
AUC Values After the Ablation Study for Different Datasets

Data Set	Detectors and AUC values			
	USLNet	BaseNet	DirectNet (ℓ_1)	DirectNet (ℓ_2)
Salinas	0.9352	0.9479	**0.9972**	0.9971
Pavia	0.9165	0.9725	**0.9915**	0.9904
El Segundo	0.9767	0.9478	0.9826	**0.9863**
Indiana	0.939	0.9921	0.9967	**0.9971**

the models to overfit anomalies and limits the detection performance. In contrast, DirectNet (coupled with a sliding dual-window model) avoids the abovementioned problems, where the inner window prevents potential contamination of anomalies to reconstruct HSIs with pure backgrounds. As a result, the capability of DirectNet in recognizing anomalies from HSIs is significantly enhanced, yielding higher AUC values. In addition, the reconstruction errors as the degree of anomaly of each pixel based on ℓ_1 and ℓ_2 have comparable effects in separating anomalous targets from the original HSI by the model, which reflects the robustness of DirectNet in the task of detecting anomalies.

F. Performance Analysis Between Target Size and Inner Window Size

To investigate the relationship between the inner window size and the target size, we regularly embed 2×2 anomalous targets on a background subimage, and their spatial sizes are set between 1×1 and 9×9. Among them, a background image and anomaly spectrum (building) were chosen from the Salinas dataset. Meanwhile, the abundance of all targets and the inner window size $W_{\text{in}} \times W_{\text{in}}$ were set to 0.3 and $\{1 \times 1, 3 \times 3, 5 \times 5, 7 \times 7, 9 \times 9\}$, respectively. Then, the AUC values attained by the proposed DirectNet on the nine synthetic datasets (using different inner window sizes) are reported in Table IV. We reiterate that, when W_{in} is set to 1, DirectNet degenerates into a blind-spot network with a single-window model (same as BaseNet in Section III-E).

We observed that BaseNet behaved with varying degrees of performance degradation on HSIs with target sizes larger than 4×4, and the larger the target sizes, the lower the corresponding AUC scores. Moreover, when W_{in} is small (e.g., 3 and 5), DirectNet appears to manifest a similar behavior on HSIs with relatively larger target sizes. However, as W_{in} increases, the applicability of DirectNet becomes broader, particularly when the inner window size is 9×9, DirectNet always performs optimally on HSIs containing synthetic targets of different sizes. This is a result of DirectNet coupling a sliding dual-window reconstruction model, where the inner window (which is not available for reconstructing the central pixel) mitigates the interference of anomalies with the outer window (the receptive field outside of the inner window) pixels, enforcing the model's ability to reconstruct the background samples on the one hand, and weakening its representation ability for anomalous samples on the other, thus achieving a purer background. Especially for large-sized targets, DirectNet can reliably leverage the self-supervised learning-based blind-spot framework for HAD purposes, thus extending its applications.

G. Comparison of Inference Times

The inference times of different methods on all datasets are recorded in Table V to assess the models' computational burden. Other than Auto-AD, BS^{3}LNet, and DirectNet, which were implemented based on the PyTorch 1.12.1 framework, the remaining methods were implemented with MATLAB 2019b. Compared to the traditional model-based detectors using several tricks (e.g., fractional Fourier transform, sliding dual window, and representation learning), GRX acquires the fastest computation time due to its clean design principle, but its detection effect is deemed relatively ineffective. Furthermore, because the five DL-based methods employ analogous procedures for HAD (these networks promise to be excellent background reconstruction models, and most of them adopt mean square error-based reconstruction errors as the final anomaly scores), they achieve comparable inference times on different datasets. Although the inference time of the proposed DirectNet is nonoptimal, DirectNet obtains promising results and its competitiveness is outstanding.

IV. Discussion

At present, mainstream DL-based HAD methods, such as GAED, RGAE, and Auto-AD, commonly adopt AE as the background reconstruction models and train the networks in an unsupervised learning manner, with the reconstruction error as the final detection result. However, the above methods may fit anomalous targets to varying degrees and yield unsatisfactory effects. In this article, we incorporate a blind-spot network based on self-supervised learning as the backbone and seamlessly embed a sliding dual-window model into this. Unlike BS^{3}LNet that concerns one-pixel blind spots, our DirectNet places the center block of the receptive field as a blind area (inner window), preventing potential contamination from anomalous pixels. Therefore, DirectNet is considered as a superior background reconstructor to enhance the background reconstruction and restrain the anomaly reconstruction. BS^{3}LNet is only applicable to recognize anomalous targets for small-sized subpixels, while DirectNet is flexible in detecting both large- and small-sized targets, expanding the application range of blind spot networks.

From a practical application perspective, our DirectNet is similar to the traditional unsupervised reconstruction model, only requiring unlabeled observed HSI to identify anomalous targets by using the reconstruction error. Yet, DirectNet can further weaken the fitting ability for anomalies, favorably separating anomalies from HSI and apparently performing better. Therefore, DirectNet will probably showcase more potential and advantages in practical applications. However, limited by the initial design concept, the insufficient generalization performance remains an existing issue for most models in the hyperspectral remote sensing field, including our DirectNet. In the HAD task, several researchers have made some efforts to achieve generalizability for models, such as WeaklyAD [81], AUD-Net [82], and AETNet [83]. Among

TABLE IV
AUC VALUES OF THE PROPOSED DIRECTNET WITH DIFFERENT PARAMETERS ON THE SYNTHETIC DATASET

Inner window size	Target size								
	1×1	2×2	3×3	4×4	5×5	6×6	7×7	8×8	9×9
1×1	1.0000	0.9992	0.9973	0.9899	0.9684	0.9060	0.8723	0.7574	0.7461
3×3	1.0000	1.0000	0.9994	0.9956	0.9860	0.9769	0.9552	0.9468	0.9084
5×5	1.0000	1.0000	0.9997	0.9924	0.9893	0.9858	0.9886	0.9869	0.9892
7×7	1.0000	1.0000	1.0000	0.9988	0.9938	0.9906	0.9846	0.9823	0.9795
9×9	1.0000	1.0000	1.0000	0.9995	0.9981	0.9952	0.9992	0.9941	0.9899

TABLE V
RUNNING TIMES (IN SECONDS) OF THE 11 CONSIDERED DETECTORS

Data Set	Detectors and running times										
	GRX	FrFE	LRX	2S-GLRT	CRD	LSDM-MoG	GAED	RGAE	Auto-AD	BS^{3}LNet	DirectNet
Salinas	0.0844	15.2690	37.4267	37.8787	2.3189	37.4213	0.1264	0.0462	**0.0289**	1.7108	1.1621
Pavia	0.0486	11.7432	17.4160	22.7698	2.9411	21.9388	0.0914	0.0307	**0.0299**	1.3722	1.2273
El Segundo	0.0621	13.0985	30.6879	45.6938	3.7150	25.4993	0.0526	0.0322	**0.0209**	1.3623	1.1594
Indiana	0.0891	23.8665	36.4680	34.8080	3.0655	8.8276	0.1003	0.0412	**0.0329**	1.4405	1.2267

them, WeaklyAD and AETNet are devoted to enhancing the discrimination between the background and anomaly of the reconstructed image and employ GRX or other detectors directly on the reconstructed image to identify anomalous targets efficiently. AUD-Net, on the other hand, aims at estimating the central pixel by the contextual information and exploits the relation embedding difference between each pixel and the surrounding pixels as the anomaly scores. Meanwhile, AUD-Net and AETNet use a huge number of training samples from multiple HSIs to improve the generalization capability of models, benefiting from the rich information provided. Unlike the aforementioned methods, our DirectNet is trained on a single HSI in a fully self-supervised manner, so as to obtain a model that can adequately reconstruct the background of this HSI, and it employs the reconstruction error to detect anomalies. The generalization performance of DirectNet is relatively limited; therefore, in future work, we will focus on designing frameworks with strong generalization capability for HAD for practical engineering applications.

V. CONCLUSION

In this article, the sliding dual-window model is creatively coupled with a deep reconstruction network for HAD purposes. To be specific, we built a background reconstruction network whose receptive field is adaptively matched to the training patch size so that the spatial–spectral features of the HSIs come into full play. Besides, the center block information of the receptive field is erased so that it is not seen by the network at all, and it is regarded as an inner window. An elaborate mask-learning strategy allows the network to concentrate on the reconstruction of the central pixel. Ultimately, our DirectNet divides its receptive field into a dual window and only uses outer window (the receptive field outside the inner window) pixels to reconstruct each pixel. As a result, the interference of anomalous samples on the reconstructed HSI is reduced, rendering the proposed DirectNet as a promising model for background reconstruction. Extensive experiments have proven the superior and comprehensive performance of our DirectNet on HSIs with anomalous targets of various sizes.

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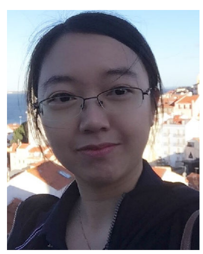

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